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Chemistry Paper 3B WASSCE (SC), 2023

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Question 1B

E is a solution containing 2.92 g of HCl per dm^3 .

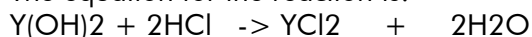
F is a solution obtained by diluting 20.0 cm^3 of a saturated solution of $\text{Y}(\text{OH})_2$ at 25 °C per dm^3 of solution.

- a. Put E into the burette and titrate it against 20.0 cm^3 or 25.0 cm^3 portion of using phenolphthalein as indicator.

Repeat the titration to obtain concordant titre values.

Tabulate your results and calculate the average volume of acid used.

The equation for the reaction is:



- b. From your results and the information provided, calculate the:

- Concentration of HCl in E mol dm^{-3} ;
- concentration of $\text{Y}(\text{OH})_2$ in F in mol dm^{-3} ;
- solubility of the substance $\text{Y}(\text{OH})_2$ in mol dm^{-3} ;
- mass of $\text{Y}(\text{OH})_2$ that would be deposited if 1 dm^3 of saturated solution is evaporated to dryness.

[H = 1.0; O = 16.0; Cl = 35.5;

$\text{Y}(\text{OH})_2 = 74.0$]

Observation

This question was on titration experiment. Majority of the candidates that responded to this question performed above average.

In part (a), majority of the candidates obtained concordant values from the titration experiment;

In part (b), majority of the candidates were able to calculate the concentration of HCl in E in mol dm^{-3} . However, they could not calculate the solubility of the

substance $Y(OH)_2$ in mol dm^{-3} .

The expected answers include:

(a) Two concordant titres

Averaging

(b) (i) Molar mass of $HCl = 1 + 35.5$

$= 36.5 \text{ g mol}^{-1}$

Concentration of E = 2.92

36.5×1

$= 0.080 \text{ mol dm}^{-3}$

(ii) From the equation

$\frac{CEVE}{CFVF} = \frac{2}{1}$

$\frac{0.080 \times VE}{CF \times VF} = \frac{2}{1}$

$CF = \frac{0.080 \times VE}{2 \times VF}$

$CF = \frac{0.080 \times VE}{2 \times VF}$

$= \text{Say } W \text{ mol dm}^{-3}$

(iii) $C_1V_1(\text{Saturated solution}) = C_2V_2(\text{dilute solution})$

$C_1 \times 20 = W \times 1000$

$C_1 = \frac{W \times 1000}{20}$

$= \text{Say } X \text{ mol dm}^{-3}$

Alternative

20cm³ of saturated solution contains W moles

Hence 1000 cm³ of saturated solution = $W \times 1000$

20

$= \text{Say } X \text{ mol dm}^{-3}$

(iv) Mass of $Y(OH)_2$ deposited

$= X \times \text{molar mass } (Y(OH)_2)$

$= X \times 74 \text{ g}$

$= Z \text{ g say}$

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